REAR-IR1: rear docking IR board, 1 3.3 V, 2 IR_L (PC8, TIM3_CH3), 3 IR_R (PC9, TIM3_CH4), 4 GND.
HALL-BIN1 / HALL-TOWER1: Hall boards, SH 3-pin (ZX-SH1.0-3PLT, so a Hall cable cannot go into a 4-pin socket): 1 3.3 V, 2 OUT (PE4 / PE5, low = magnet near), 3 GND. All mounting pads to GND.

CLIFF
CLIFF-LMD
CLIFF-RMD
CLIFF_LED_EN
CLIFF-RRD
CLIFF-RFD
CLIFF-LRD
CLIFF-LFD
File: CLIFF.kicad_sch

MAIN BRUSH MOTOR
MAIN-BRUSH-IN1
MAIN-BRUSH-IN2
MAIN-BRUSH-ADC
File: MAIN-BRUSH-MOTOR .kicad_sch

WHEEL MOTORs
WHEEL-MOT-R-IN2
WHEEL-MOT-R-IN1
WHEEL-MOT-R-ENCODED
WHEEL-MOT-R-DROPD
WHEEL-MOT-L-IN2
WHEEL-MOT-L-IN1
WHEEL-MOT-L-DROPD
WHEEL-MOT-L-ENCODED
WHEEL-MOT-R-ADCD
WHEEL-MOT-L-ADCD
File: WHEEL-MOTORs .kicad_sch

MAIN FAN
MAIN-FAN-FBD
MAIN-FAN-PWMD
File: MAIN-FAN .kicad_sch

SIDE BRUSH MOTORs
SIDE-BRUSH-IN1
SIDE-BRUSH-IN2
SIDE-BRUSH-ADCD
File: SIDE-BRUSH-MOTORs .kicad_sch

Bumper Sensors
BUMPER
BUMP-L
BUMP-R
BUMP_L_LED_EN
BUMP_R_LED_EN
File: bumper_sensors.kicad_sch

VSENS-3V3
SIDE-R1
ZX-SH1_0-7PWT
USART2-IR-R-RX
SIDE-PROXI-CE-R
I2C4_SCL
I2C4_SDA

MCU-STM32
CLIFF-LF
CLIFF-LR
CLIFF-RF
CLIFF-RR
CLIFF_LED_END
CLIFF-RM
CLIFF-LM
BUMP-L
BUMP-R
BUMP_L_LED_END
BUMP_R_LED_END
USART2-IR-R-RX
SIDE-PROXI-CE-RD
I2C4_SDA
I2C4_SCLD
CARPET-SENSOR-LOD
CARPET-SENSOR-ECHO
WDID
LIDAR_RXD
LIDAR_TX
LIDAR-MOTOR-CTRLD
INT-SENSORS
VL-L-LPnD
VL-I2C-RESETD
UART3_RX2
UART3_RX1
DP
DN
USB_CC1D
USB_CC2
USB_VBUS_SENSE
USB_PWR_END
VMOT-14V4-END
REAR-IR-L
REAR-IR-R
IMU-INT#1
LED-HOME
LED-PWR
CLEAN-BTN
HOME-BTN
BAT_ID
DC_IN_SENSE
BAT_SENSE
VSENS_5V_USB_EN
VSENS_3V3_EN
PWR-BTN-SENSE
STM-PWR-CTRL
File: STM32G473.kicad_sch

VSENS-3V3
R61
47k
INT-SENSORS
VL-L-LPn
VL-I2C-RESET

CARPET-SENSOR
LO
ECHO_ADCD
File: Carpet-sensor.kicad_sch

WATCHDOG
WD
WDI
WD_OKD
File: WATCHDOG.kicad_sch

WD_OK

LiDAR
WD_OK
LiDAR-RXD
LiDAR-TXD
LiDAR-MOTOR-CTRL
LiDAR-EN
File: LIDAR.kicad_sch

FRONT SENSORS
INT-SENSORS
VL-L-LPn
VL-I2C-RESET
IR-F-RX2
IR-F-RX1
VL-I2C-SCL
VL-I2C-SDA
File: Front-Sensors.kicad_sch

USB Power
USB-C POWER
DP
DN
USB_CC1
USB_CC2
USB_VBUS_SENSE
USB_PWR_EN
File: usb_power.kicad_sch

HALL-BIN1 VSENS-3V3
ZX-SH1_0-3PLT
HALL-TOWER1 VSENS-3V3
ZX-SH1_0-3PLT
REAR-IR1 VSENS-3V3
REAR-IR-L
REAR-IR-R
ZX-SH1_0-4PLT
HALL-BIN
HALL-TOWER

VSENS-3V3
R6 2.2k
R5 2.2k

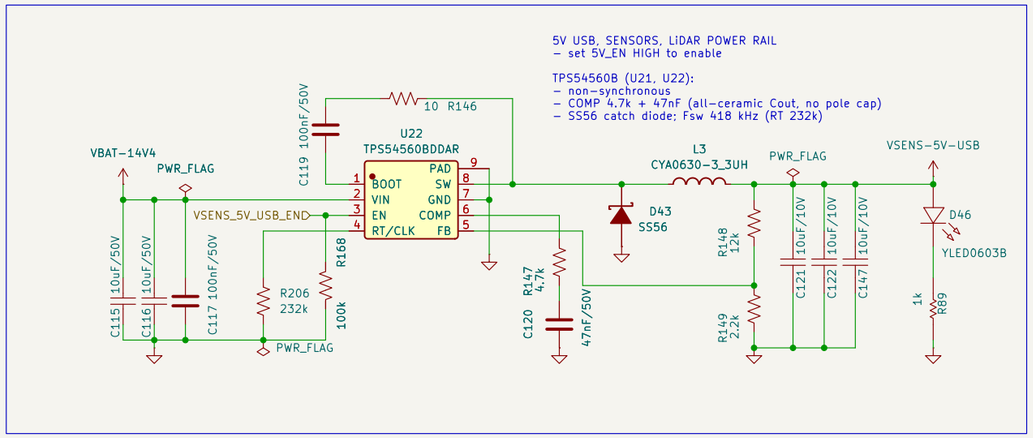
IMU
IMU-INT#1D
IMU-SDA
IMU-SCL
File: IMU-ICM-4267-P .kicad_sch

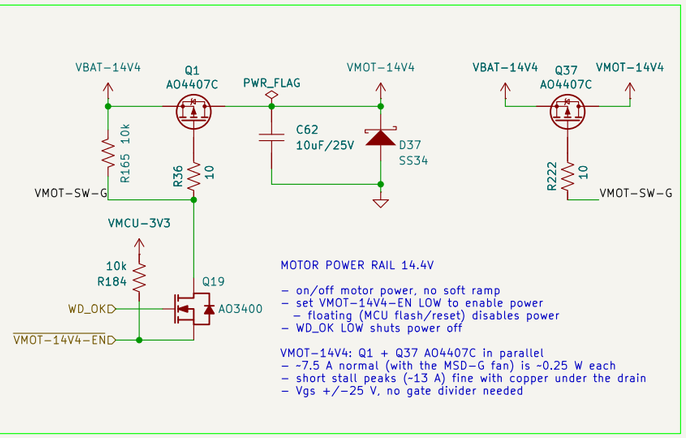
POWER CONTROL
VMOT-14V4-EN
STM-PWR-CTRL
PWR-BTN-SENSE
VSENS_3V3_EN
VSENS_5V_USB_EN
WD_OK
PWR-SW-G
POWER-BTN
File: SYSTEM-POWER.kicad_sch

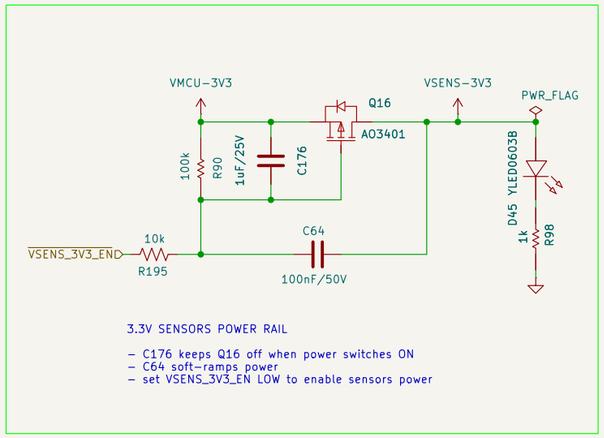
BUTTON-LEDs
POWER-BTND
LED-HOME
LED-PWR
CLEAN-BTND
HOME-BTND
File: BUTTON-LEDs .kicad_sch

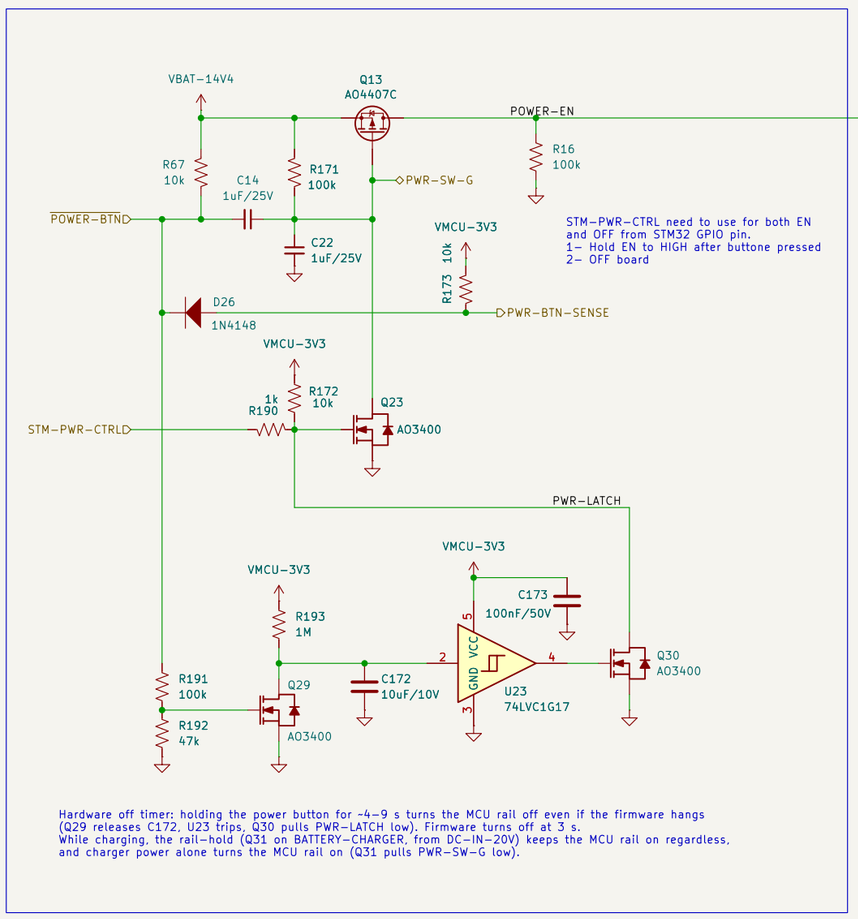
BATTERY CHARGER
BATTERY-CHARGER
PWR-SW-G
BAT_SENSE
DC_IN_SENSE
ID
File: Battery-Charger.kicad_sch

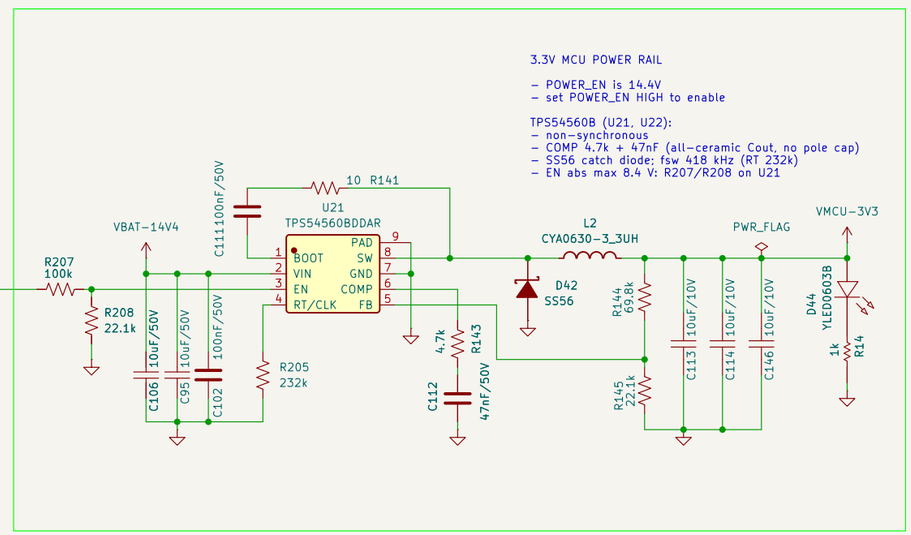
Sheet: /
File: Main.kicad_sch
Title:
Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 1/17

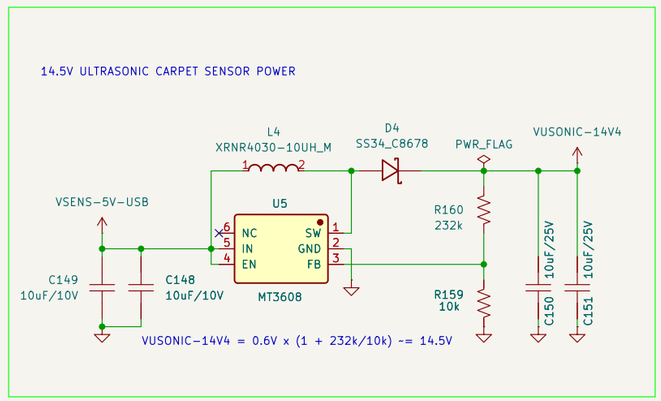


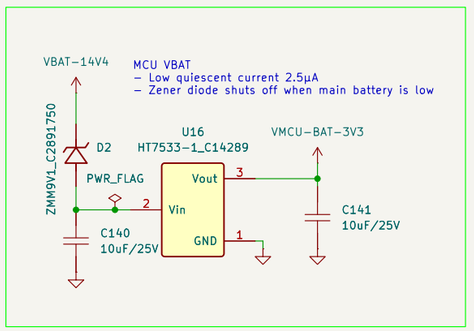






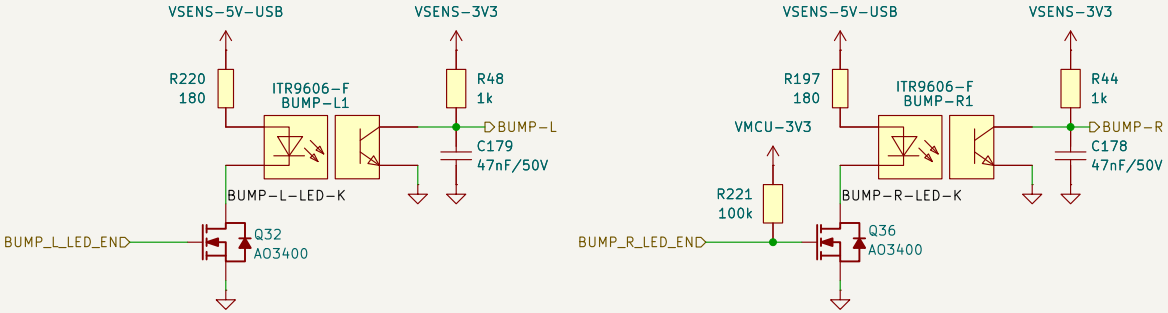






MP1 MP2 MP3 MP4 MP5 MP6

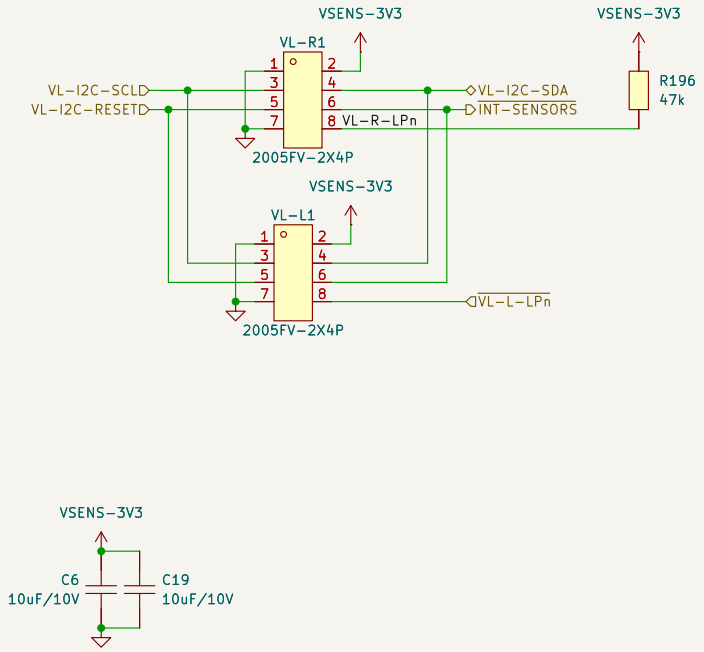
Sheet: /POWER/		
File: SYSTEM–POWER.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 2/17



- slotted photo-interrupters
- BUMP-X reads low when light passes the slot
- drive BUMP_L/R_LED_EN PWM <<15 usec period to enable
- tune R44, R48, LED PWM duty to let ADC sense full open/close range
  1k keeps photo transistor output out of saturation
  so PA4/PA5 (ADC2) see the transition (ITR9606 IC(ON) 0.5-10 mA)
- R221 enables MCU USB-C Rd on both CC1 and CC2 during MCU *reset*,
  so the phone detects the robot whichever way the cable is plugged in

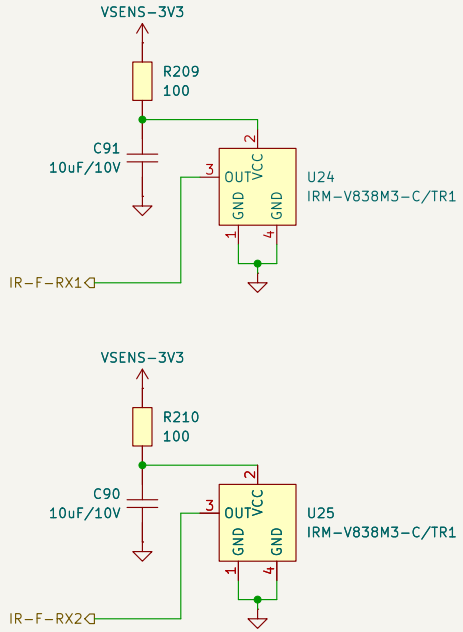
Sheet: /Bumper Sensors/	
File: bumper_sensors.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 3/17

2x VL53L7 ToF BOARDS

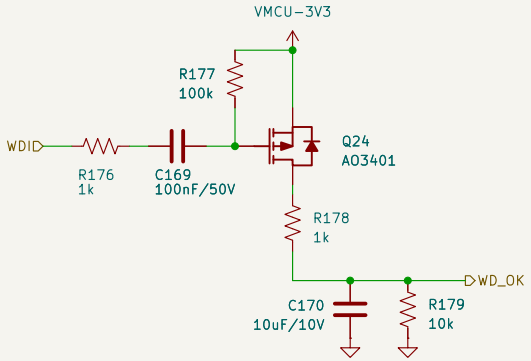


2 IR RECEIVERS

U24 left, U25 right (robot view). Side-looking: put both at the board front edge, lenses facing forward, ~10 mm apart, with a baffle between them in the bumper window. – 90 deg FoV



Sheet: /FRONT SENSORS/	
File: Front-Sensors.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 4/17



RC CHARGE-PUMP WATCHDOG

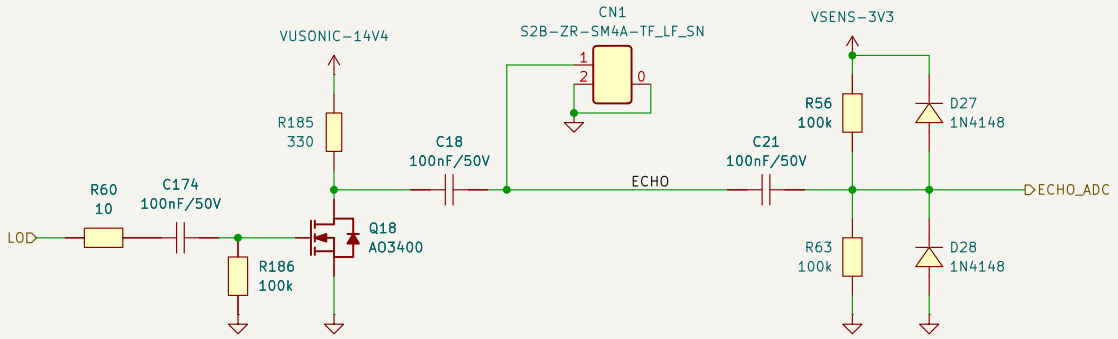
- WD_OK active HIGH
- WDI HIGH->LOW transition pets the dog

Firmware toggles WDI in software (50-1000 Hz, ~50% duty) only while the control loop is healthy.
Never drive WDI from a timer/PWM: a timer keeps running after the CPU hangs.
Each falling edge on WDI pulls the Q24 gate low through C169 and tops up C170 via R178.
WD_OK ~2.7 V while toggling. WDI stuck high, low or floating: <1 V in ~100 ms, <0.4 V in ~190 ms
(10 uF nominal; DC-bias derating shortens this). WD_OK starts at 0 V at power-up (fail-safe).
WD_OK drives N-FET gates only (14.4 V rail switch, LiDAR incl. laser): no resistive loads.
WDI pin: high-Z at reset, no pull-down.

Sheet: /WATCHDOG/	
File: WATCHDOG.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	Id: 5/17

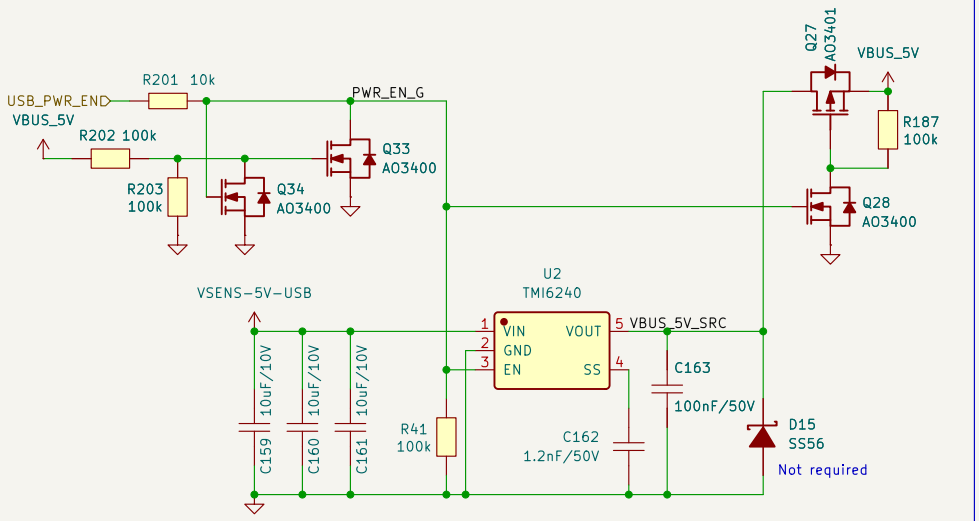
Carpet sensor driver: one low-side AO3400 + 330R pull-up from VUSONIC-14V4 (14.4 V). The P-FET half-bridge was removed: its 100k gate pull-up could not turn it off at 290 kHz (shoot-through). R186 holds LO low in reset/flash.
Drive LO in short bursts only: LO stuck high puts 0.63 W into R185.

C174 AC-couples LO to the Q18 gate: if LO sticks high, the gate decays through R186 (~10 ms) and Q18 turns off, so R185 is never stressed by firmware holding LO high.



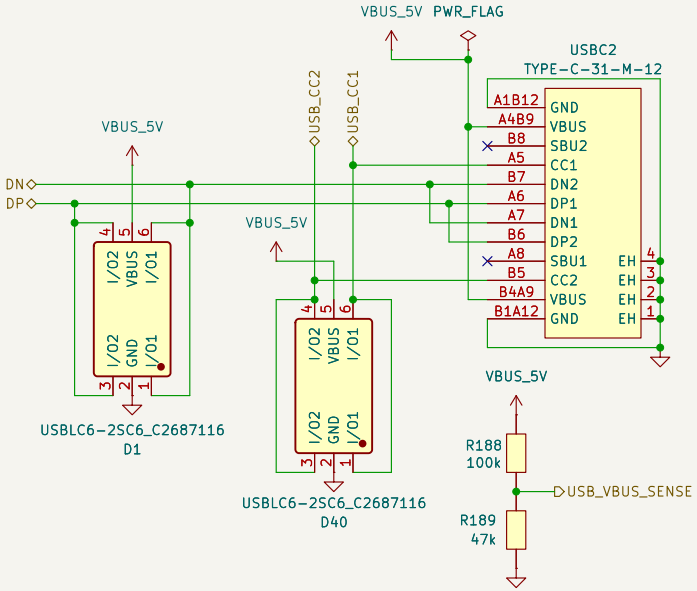
Sheet: /CARPET-SENSOR/		
File: Carpet-sensor.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/17

USB-C for SMARTPHONE



VBUS_5V lockout: the robot 5 V (TMI6240 + Q27) can only turn on while VBUS_5V is ~0 V. Q33 holds PWR_EN_G low if a phone already sources VBUS_5V; once on, Q34 latches it.

PD power-role swap lets the robot charge the phone while it stays host. Q27 blocks the phone's VBUS_5V from back-feeding the robot 5 V (the TMI6240 body diode conducts VOUT->VIN); USB_PWR_EN turns on TMI6240 and Q27 together.
TMI6240 now runs from VSENS-5V-USB (U25 buck removed).

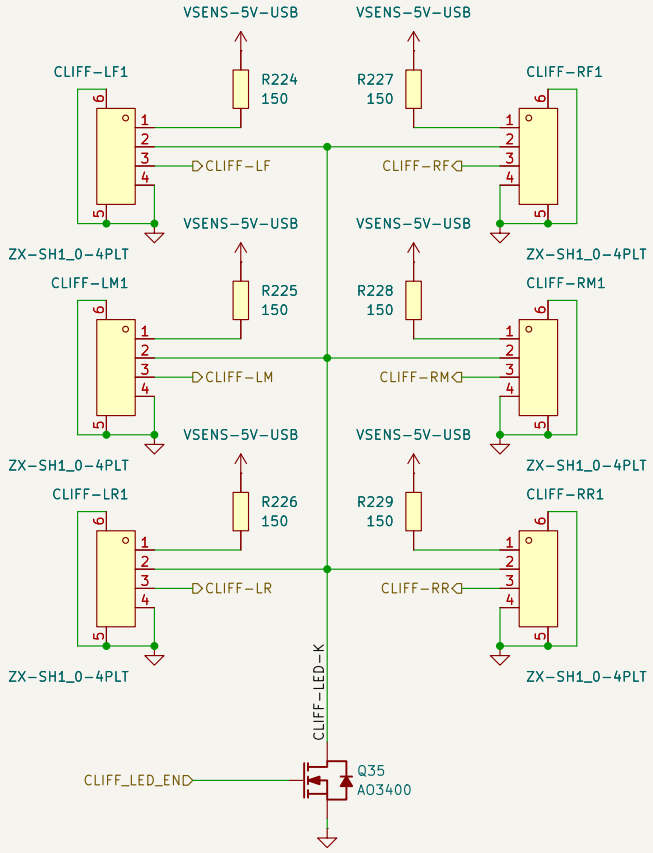


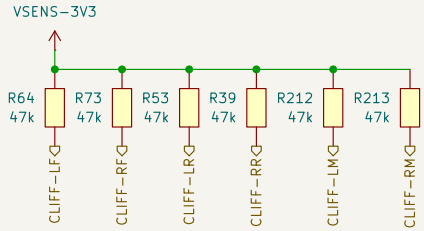
USB-C (phone): phone is the USB host. CC1/CC2 go to the STM32 UCPD (PB6/PB4): Rd by default.

Sheet: /USB Power/
File: usb_power.kicad_sch

Title:

Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 7/17



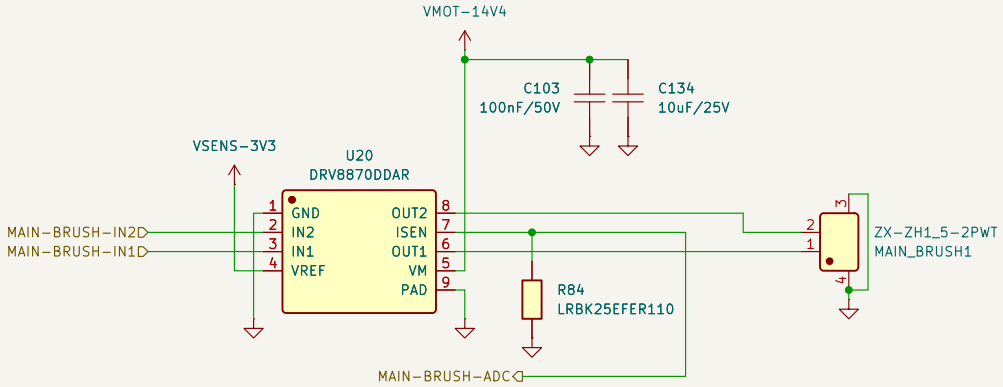


- kicad/cliff-sensor mating boards
- SH 4-pin vertical; use same (pin 1 to pin 1) cable
- set CLIFF_LED_EN LOW to disable
- modulate CLIFF_LED_EN, read each channel LED-on minus LED-off to cancel ambient light
- possible TODO - cliff LEDs 2x or 3x in series, save resistors
  - worsens power rail noise rejection
  - ADC fw could reject noise by correlating the 6 readings
- 0.76W power consumption when enabled continuously
  - drive at ~10% duty cycle to save power
- tweak output pullup
  - a low resistor value can saturating the photo transistor
  - 47K pullup makes for 1ms output settling time
  - can use 10k for faster settling, to reduce duty cycle further
- combine low duty with fast duty LED cycling
  - fast (~10x photo transistor settling time) reduces LED average radiance to avoid photo transistor saturation
- check sensor faults - is output 3.3V or floating?
  - normal no-cliff condition - photo transisstor gets LED light Vary LED current and check photo-transistor voltage.

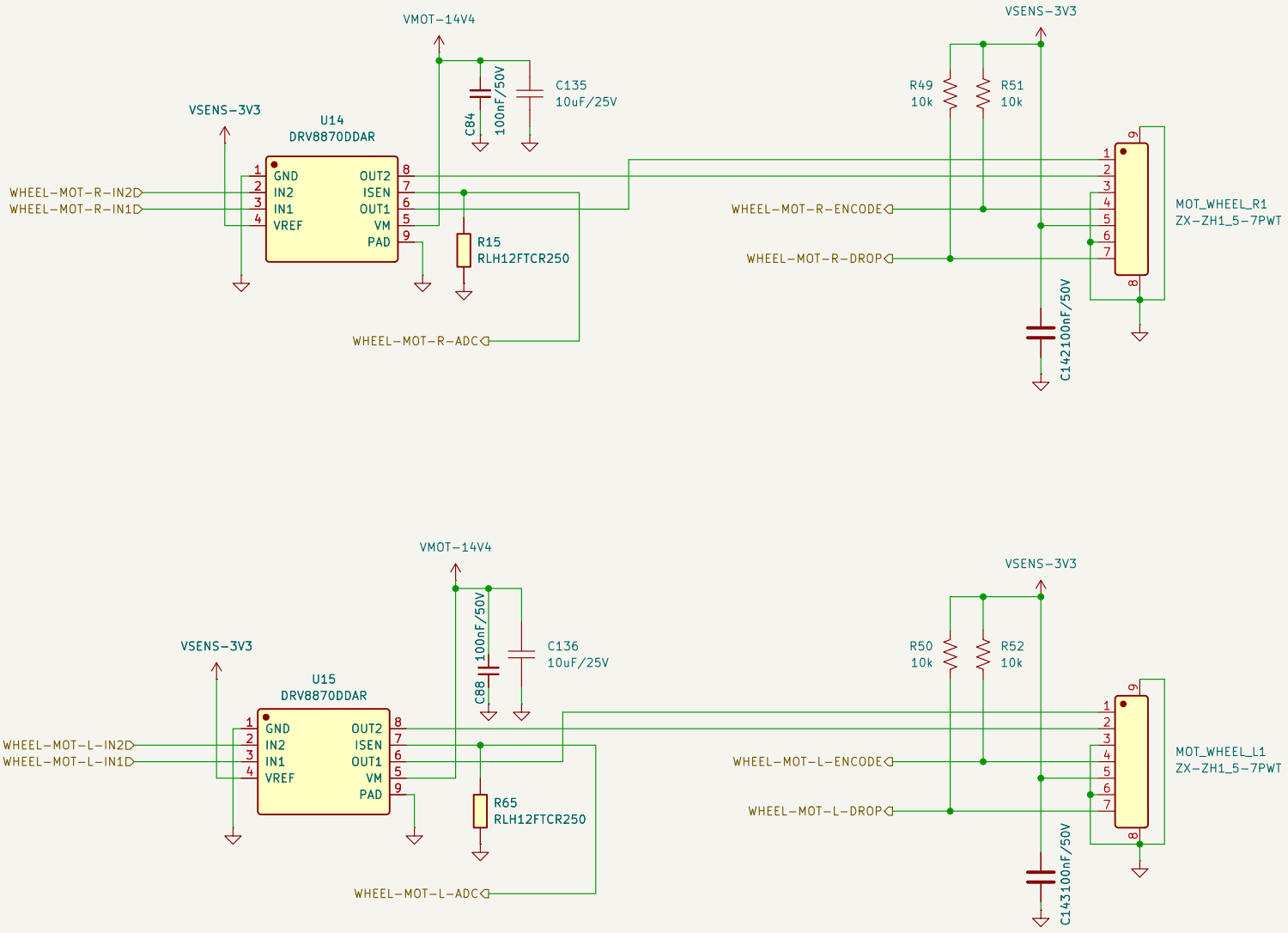
Sheet: /CLIFF/
File: CLIFF.kicad_sch

Title:

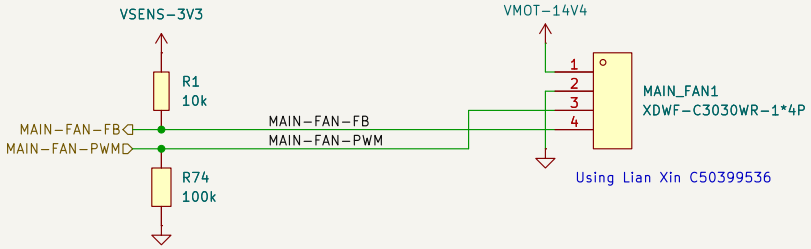
Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 19/17



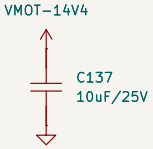
Sheet: /MAIN-BRUSH MOTOR/		
File: MAIN-BRUSH-MOTOR .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 22/17



Sheet: /WHEEL MOTOR/	
File: WHEEL-MOTORs .kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	Id: 23/17



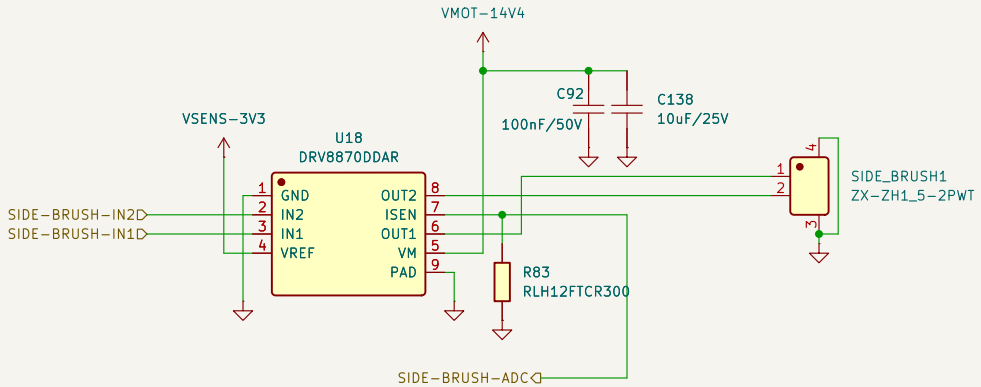
- FB feedback signal is open collector
- set PWM LOW to turn fan OFF
- MAIN_FAN1 matches Dreame L60 PE Ultra PCB



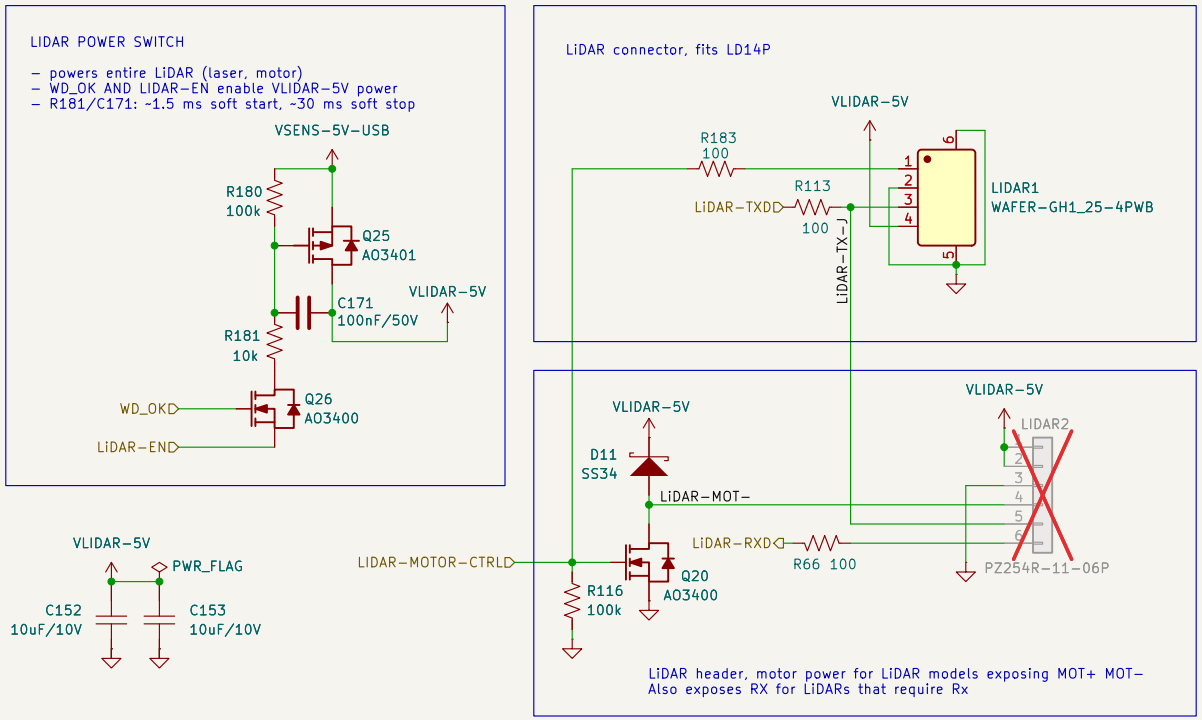
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File: MAIN-FAN .kicad_sch

Title:

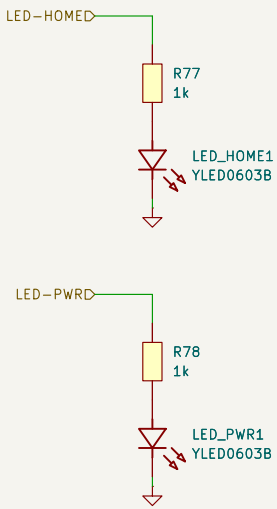
Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 24/17

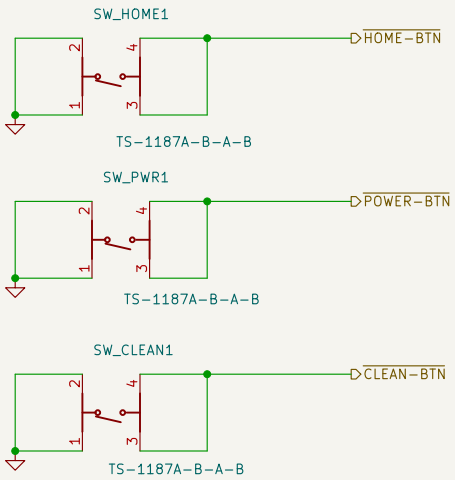


Sheet: /SIDE-BRUSH/		
File: SIDE-BRUSH-MOTORs .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 25/17

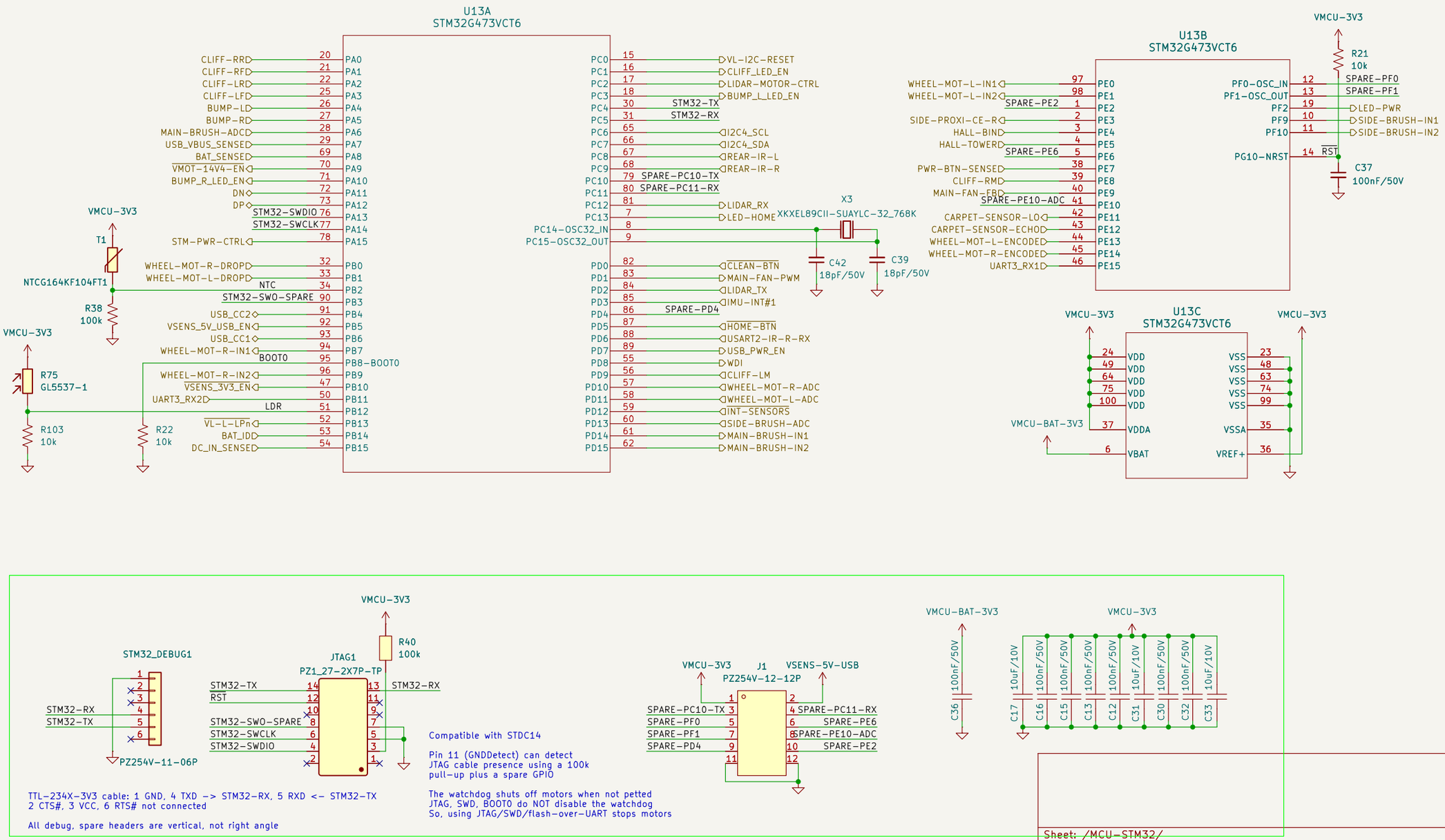


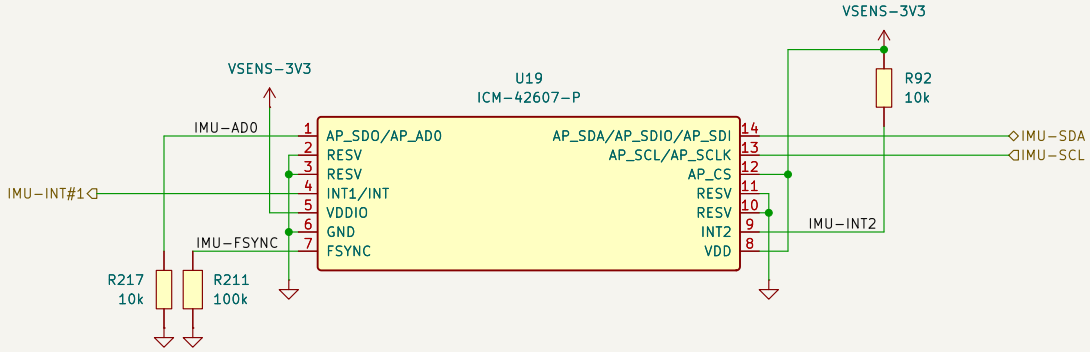
Sheet: /LIDAR/	
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Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 26/17



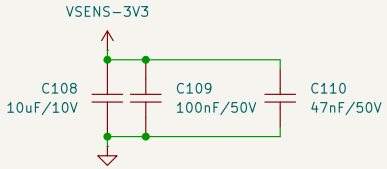


Sheet: /BUTTON-LED/ File: BUTTON-LEDs .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 28/17


U13A
STM32G473VCT6
CLIFF-RR
CLIFF-RF
CLIFF-LR
CLIFF-LF
BUMP-LD
BUMP-RD
MAIN-BRUSH-ADC
USB_VBUS_SENSED
BAT_SENSED
VMOT-14V4-EN
BUMP_R_LED_EN
DN
DP
STM32-SWDIO
STM32-SWCLK
STM-PWR-CTRL
WHEEL-MOT-R-DROP
WHEEL-MOT-L-DROP
NTC
STM32-SWO-SPARE
USB_CC2
VSENS_5V_USB_EN
USB_CC1
WHEEL-MOT-R-IN1
BOOT0
WHEEL-MOT-R-IN2
VSENS_3V3_EN
UART3_RX2
LDR
VL-L-LPn
BAT_ID
DC_IN_SENSED
VL-I2C-RESET
CLIFF_LED_EN
LIDAR-MOTOR-CTRL
BUMP_L_LED_EN
STM32-TX
STM32-RX
I2C4_SCL
I2C4_SDA
REAR-IR-L
REAR-IR-R
SPARE-PC10-TX
SPARE-PC11-RX
LIDAR_RX
LED-HOME
PC14-OSC32_IN
PC15-OSC32_OUT
CLEAN-BTN
MAIN-FAN-PWM
LIDAR_TX
IMU-INT#1
SPARE-PD4
HOME-BTN
USART2-IR-R-RX
USB_PWR_EN
WDI
CLIFF-LM
WHEEL-MOT-R-ADC
WHEEL-MOT-L-ADC
INT-SENSORS
SIDE-BRUSH-ADC
MAIN-BRUSH-IN1
MAIN-BRUSH-IN2
VMCU-3V3
T1
NTCG164KF104FT1
R38
100k
R75
GL5537-1
R103
10k
R22
10k
X3
XKXEL89CII-SUAYLC-32_768K
C42
18pF/50V
C39
18pF/50V
U13B
STM32G473VCT6
WHEEL-MOT-L-IN1
WHEEL-MOT-L-IN2
SPARE-PE2
SIDE-PROXI-CE-R
HALL-BIN
HALL-TOWER
SPARE-PE6
PWR-BTN-SENSED
CLIFF-RM
MAIN-FAN-FB
SPARE-PE10-ADC
CARPET-SENSOR-LO
CARPET-SENSOR-ECHO
WHEEL-MOT-L-ENCODED
WHEEL-MOT-R-ENCODED
UART3_RX1
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
PF0-OSC_IN
PF1-OSC_OUT
PF2
PF9
PF10
PG10-NRST
SPARE-PF0
SPARE-PF1
LED-PWR
SIDE-BRUSH-IN1
SIDE-BRUSH-IN2
RST
R21
10k
C37
100nF/50V
U13C
STM32G473VCT6
VDD
VDDA
VBAT
VSS
VSSA
VREF+
VMCU-BAT-3V3
STM32_DEBUG1
PZ254V-11-06P
JTAG1
PZ1_27-2X7P-TP
R40
100k
Compatible with STDC14
Pin 11 (GNDDetect) can detect
JTAG cable presence using a 100k
pull-up plus a spare GPIO
The watchdog shuts off motors when not petted
JTAG, SWD, BOOT0 do NOT disable the watchdog
So, using JTAG/SWD/flash-over-UART stops motors
J1
PZ254V-12-12P
SPARE-PC10-TX
SPARE-PF0
SPARE-PF1
SPARE-PD4
SPARE-PC11-RX
SPARE-PE6
SPARE-PE10-ADC
SPARE-PE2
C36
C17
C16
C15
C13
C12
C31
C30
C32
C33
100nF/50V
10uF/10V
TTL-234X-3V3 cable: 1 GND, 4 TXD -> STM32-RX, 5 RXD <- STM32-TX
2 CTS#, 3 VCC, 6 RTS# not connected
All debug, spare headers are vertical, not right angle
Sheet: /MCU-STM32/
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IMU on I2C4 (2.2k pull-ups R5/R6 on the top sheet). AP_AD0 pulled low by R217 (I2C address 0x68).
The IMU is leadless: probe AD0 on R217, INT2 on its pull-up R92 and FSYNC on R211. Run a wire to a spare pin if INT2 or FSYNC is needed.

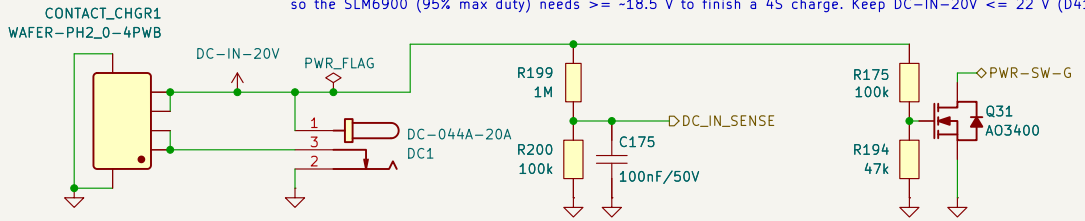


Sheet: /IMU/
File: IMU-ICM-4267-P .kicad_sch

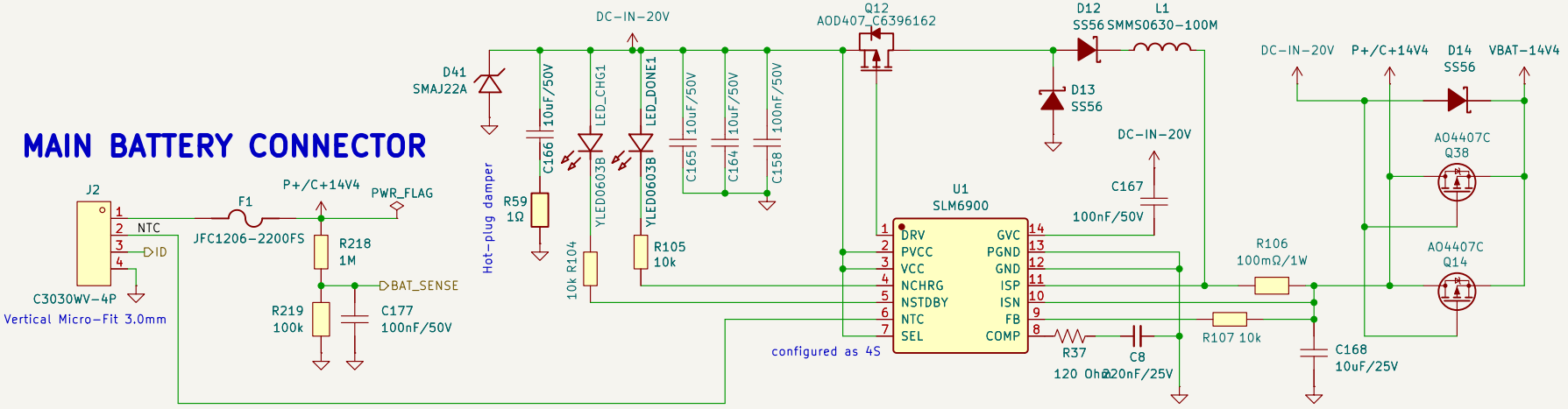
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KiCad E.D.A. 10.0.4 | Id: 32/17

DC-IN-20V: 20 V brick or dock. With the battery disconnected (Q14 off) VBAT-14V4 = DC-IN-20V - 0.4 V,
so the SLM6900 (95% max duty) needs >= ~18.5 V to finish a 4S charge. Keep DC-IN-20V <= 22 V (D41, Q13 Vgs).



MAIN BATTERY CONNECTOR



Rail-hold: charger power (dock or barrel jack)
turns the MCU rail on and keeps it on.
DC-IN-20V to GND ~130k (R175+R194 || R199+R200)
= dock presence signature.

R106 sets charging current
100 mOhm 1.2 A

Sheet: /BATTERY-CHARGER/	
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Size: USLetter | Date:	**Rev:**
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